Unit 5 scientist comparison chart

public-student | All competition levels

The equations are modern associations, not a claim that each individual wrote the modern formula alone. Life dates provide context; contribution dates belong in the timeline.

Scientist and lifespan	Contribution	Modern concept and significance
Joseph Black (1728–1799)	Specific heat; latent heat	Separated temperature from thermal quantity; foundation for calorimetry. $Q=mc\Delta T$; $Q=mL$ (modern notation) [H01,H02]
Sadi Carnot (1796–1832)	Ideal reversible engine reasoning	Established roots of a general performance limit; original caloric framework later revised. Modern $\eta_{max}=1-T_C/T_H$ [H06]
James Prescott Joule (1818–1889)	Mechanical equivalent of heat	Quantitative experimental support for energy conversion and conservation. $\Delta U=Q-W$; joule J [H05]
Rudolf Clausius (1822–1888)	Second-law development; entropy	Added direction and irreversibility to energy accounting. Total isolated entropy cannot decrease [H07]
William Thomson / Lord Kelvin (1824–1907)	Absolute temperature; thermodynamic theory	Developed a general temperature framework; original proposal later refined. $T=t+273.15$; Kelvin ratios [H03,H04]
James Clerk Maxwell (1831–1879)	Molecular velocity distribution; demon	Connected particle populations with bulk properties; examined second-law assumptions. Statistical gas description [H10,H11,H14]
Ludwig Boltzmann (1844–1906)	Statistical interpretation of entropy	Explained thermal regularities through microscopic possibilities. Multiplicity; k_B links energy and K [H12,H13,H04]
Max Planck (1858–1947)	Radiation law; energy quanta	Opened quantum theory through the thermal-radiation problem. $\varepsilon=hf$ conceptually [H08,H09]

Three attribution checks

Clausius named entropy; Boltzmann developed its statistical interpretation. Kelvin’s scale concerns absolute temperature; Black’s latent heat concerns energy during phase change. Joule’s conservation evidence and Carnot’s ideal limits answer different questions.